

# Notes: Fang, Liu, and Roussanov (RFS, 2026)

## Getting to the Core: Inflation Risks Within and Across Asset Classes

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# 1 Big picture

## 1.1 Summary

The paper shows that **core inflation** (not energy inflation) is the uniquely and robustly **priced** inflation risk across many asset classes: most assets have *negative* core-inflation betas and thus earn positive premia because the *price of core inflation risk* is negative; energy inflation exposures are important for hedging but are not stably priced.

## 1.2 Why the multi-asset-class design is essential

A single asset class (e.g., only equities) provides limited cross-sectional dispersion in inflation betas and returns. By stacking portfolios from *eight* asset classes, the paper creates large variation in exposures and average returns, which strengthens identification of the inflation risk premium and allows “within vs across” consistency checks.

## 1.3 Core empirical objects

The empirical analysis revolves around four linked objects:

- (a) **Inflation shocks** extracted from a reduced-form VAR (headline, core, food, energy).
- (b) **Inflation betas** of asset returns (inflation hedging regressions).
- (c) **Prices of inflation risks** (cross-sectional pricing of beta differences).
- (d) **Mechanisms** (growth predictability and high-frequency announcement responses).

## 1.4 Structural interpretation

A two-sector New Keynesian (NK) asset pricing model is constructed to match: (i) the negative price of *core* inflation risk, (ii) distinct signs of core vs energy betas across asset classes, and (iii) time variation in the stock–bond correlation via changing energy-demand dynamics and policy responses.

## 2 Data, inflation measurement, and notation

### 2.1 Inflation series

Let

$$\pi_t^h \equiv \text{headline inflation}, \quad \pi_t^c \equiv \text{core inflation}, \quad \pi_t^f \equiv \text{food inflation}, \quad \pi_t^e \equiv \text{energy inflation}.$$

All inflation rates are annualized. Table 1 in the paper summarizes means, volatilities, persistence, and correlations.

### 2.2 Asset universe and test portfolios

The paper studies returns across **eight asset classes**:

- (1) U.S. equities (industry portfolios),
- (2) U.S. Treasuries (maturity portfolios),
- (3) Agency bonds,
- (4) Corporate bonds,
- (5) Currencies (carry portfolios plus dollar-carry),
- (6) Commodity futures (sector portfolios),
- (7) REITs (equity/mortgage/hybrid),
- (8) International equities (regional portfolios).

Two portfolio sets are used repeatedly:

- **“8 average portfolios”**: one portfolio per asset class (equal-weighted average across portfolios within the class).
- **“38 portfolios”**: a richer cross-section consisting of multiple portfolios per class (Table 3).

### 2.3 Return notation

Let  $R_{i,t}$  denote the nominal return on asset (or portfolio)  $i$  in period  $t$ , and let  $R_{f,t}$  denote the nominal risk-free rate. Define the **nominal excess return**

$$re_{i,t} \equiv R_{i,t} - R_{f,t}.$$

A key point in interpretation: because  $R_{f,t}$  already embeds expected inflation, regressions of  $re_{i,t}$  on inflation shocks primarily capture the asset’s reaction to *unexpected* inflation news.

### 3 Inflation shocks from a reduced-form VAR

#### 3.1 VAR specification (paper Eq. (1))

Inflation shocks are the innovations from the reduced-form VAR:

$$Y_t = c + AY_{t-1} + \varepsilon_t, \quad (1)$$

where  $Y_t$  stacks:

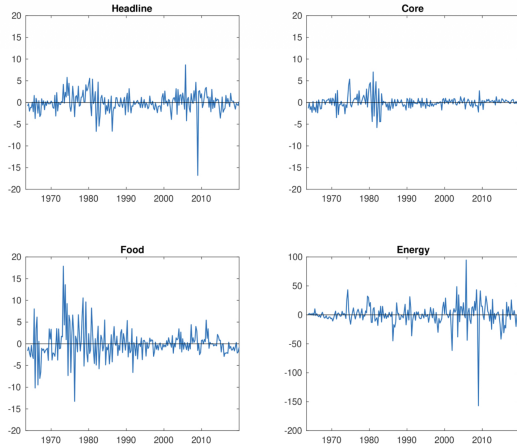
$$Y_t = (\pi_t^h, \pi_t^c, \pi_t^f, \pi_t^e, r_{f,t}, \log(PD_t), \text{output gap}_t)^\top.$$

The first four components of  $\varepsilon_t$  are taken as **inflation innovations** for headline, core, food, and energy inflation, respectively.

#### 3.2 Interpretation

- The VAR is *reduced form*: shocks are not assigned to structural sources (productivity, markups, demand, policy, etc.).
- The aim is to build **unexpected inflation news** that is orthogonal to lagged information in  $Y_{t-1}$ .

#### 3.3 Empirical inflation facts (Table 1 and Figure 1)



**Figure 1**  
**Inflation shocks**  
This figure plots the time series of headline, core, food, and energy inflation shocks extracted from VAR described in the main text. Data are quarterly from 1963Q3 to 2019Q4.

**Table 1**  
**Summary statistics of inflation**

| A. Summary     |          |       |          |        |
|----------------|----------|-------|----------|--------|
|                | Mean     | SD    | Autocorr |        |
| Headline       | 3.76     | 3.24  | 0.60     |        |
| Core           | 3.75     | 2.66  | 0.79     |        |
| Food           | 3.75     | 4.04  | 0.43     |        |
| Energy         | 4.01     | 19.52 | 0.04     |        |
| B. Regression  |          |       |          |        |
|                | $\beta$  |       | SE       |        |
| Core           | 0.71     |       | 0.01     |        |
| Food           | 0.20     |       | 0.01     |        |
| Energy         | 0.09     |       | 0.00     |        |
| C. Correlation |          |       |          |        |
|                | Headline | Core  | Food     | Energy |
| Headline       | 1.00     |       |          |        |
| Core           | 0.80     | 1.00  |          |        |
| Food           | 0.60     | 0.44  | 1.00     |        |
| Energy         | 0.69     | 0.20  | 0.17     | 1.00   |

The table provides summary statistics of headline inflation and its three components: core, food, and energy inflation. Panel A reports the mean, standard deviation, and autocorrelation of each series. Panel B reports the regression coefficients of headline inflation on core, food, and energy inflation. Panel C reports the correlation matrix. Inflation is annualized. Data are quarterly from 1963Q3 to 2019Q4.

Table 1 establishes three stylized facts that drive the entire paper:

- (1) **Similar means:** headline/core/food/energy average around  $\approx 4\%$  per year.
- (2) **Very different second moments:** core inflation is persistent and low-volatility; energy inflation is extremely volatile and weakly persistent.
- (3) **Core dominates headline:** the regression of headline inflation on components assigns the largest weight to core, consistent with the CPI basket.

Figure 1 visualizes why energy inflation is empirically special: its shock series contains large spikes compared to core.

## 4 Inflation hedging regressions: estimating betas

### 4.1 Baseline beta regression (paper Eq. (2))

The inflation-hedging regression is:

$$re_{i,t} = \alpha_i + \beta_i^\pi \varepsilon_{\pi,t} + u_{i,t}, \quad (2)$$

where  $\varepsilon_{\pi,t}$  is either (i) the headline inflation shock, or (ii) a vector of shocks  $(\varepsilon_t^c, \varepsilon_t^e)$  for core and energy.

### 4.2 How to read $\beta_i$

- $\beta_i^\pi = 1$  would mean that asset  $i$  moves one-for-one with the inflation surprise *in excess-return space* and hence perfectly hedges unexpected inflation.
- $\beta_i^\pi < 0$  means the asset performs poorly when inflation surprises are positive (“bad inflation hedge”).
- The paper’s central empirical pattern is not about  $\beta_i^h$  alone; it is about the **sign flip between core and energy exposures** for many assets.

### 4.3 What Table 2 says (8 average portfolios)

Table 2 summarizes the *average* inflation betas by asset class. A coherent way to read it is to compare **core vs energy** columns:

- **Equities and REITs:** strongly negative *core* betas (bad hedge for core inflation) but positive *energy* betas (tend to rise with energy inflation).
- **Treasuries and agency/corporate bonds:** negative betas for both core and energy (inflation hurts bonds through expected inflation and discounting).
- **Commodities:** essentially no core exposure but strongly positive energy exposure (commodities hedge energy inflation by construction).
- **Currencies:** more mixed; carry portfolios can load negatively on U.S. core inflation and positively on energy inflation.

### 4.4 What Table 3 adds (38 portfolios)

Table 3 reveals **within-class heterogeneity**:

- Within equities, industry portfolios differ in headline betas, but *all* have negative and significant core betas.

**Table 2**  
Average portfolio exposures to inflation risks

|            | Mean | SD    | A. Headline      |         | B. Core and energy |         |                |         |
|------------|------|-------|------------------|---------|--------------------|---------|----------------|---------|
|            |      |       | headline $\beta$ | t-stat  | core $\beta$       | t-stat  | energy $\beta$ | t-stat  |
| Stock      | 6.80 | 16.79 | -1.33            | (-1.38) | -5.60              | (-3.69) | 0.21           | (1.81)  |
| Treasury   | 2.07 | 6.90  | -2.53            | (-7.06) | -2.51              | (-4.27) | -0.20          | (-4.57) |
| Agency     | 2.44 | 5.10  | -1.62            | (-5.42) | -2.25              | (-4.28) | -0.09          | (-2.75) |
| Corporate  | 3.08 | 6.39  | -1.60            | (-4.38) | -2.98              | (-4.91) | -0.05          | (-1.08) |
| Currency   | 1.76 | 7.05  | 1.04             | (2.02)  | -1.04              | (-0.65) | 0.13           | (2.54)  |
| Commodity  | 4.47 | 21.90 | 8.59             | (7.53)  | -0.07              | (-0.04) | 1.10           | (8.21)  |
| REIT       | 7.96 | 17.46 | 0.31             | (0.27)  | -6.54              | (-3.30) | 0.31           | (2.48)  |
| Intl stock | 6.09 | 16.53 | -1.20            | (-1.23) | -5.78              | (-3.74) | 0.19           | (1.70)  |

**Table 3**  
Asset return exposure to inflation risks

|                            | Mean  | SD    | A. Headline      |         | B. Core and energy |         |                |         |
|----------------------------|-------|-------|------------------|---------|--------------------|---------|----------------|---------|
|                            |       |       | headline $\beta$ | t-stat  | core $\beta$       | t-stat  | energy $\beta$ | t-stat  |
| <i>Stock</i>               |       |       |                  |         |                    |         |                |         |
| Consumer                   | 7.83  | 17.70 | -2.62            | (-2.61) | -6.34              | (-3.97) | 0.06           | (0.48)  |
| Manufacturing              | 6.65  | 15.49 | 0.32             | (0.35)  | -4.20              | (-3.02) | 0.36           | (3.39)  |
| High Tech                  | 7.31  | 20.29 | -1.17            | (-1.00) | -6.07              | (-3.29) | 0.26           | (1.86)  |
| Health                     | 8.67  | 17.80 | -2.73            | (-2.70) | -6.30              | (-3.91) | 0.04           | (0.34)  |
| Others                     | 7.27  | 20.09 | -2.38            | (-2.08) | -7.40              | (-4.09) | 0.17           | (1.22)  |
| <i>Treasury</i>            |       |       |                  |         |                    |         |                |         |
| 1-year                     | 0.96  | 1.87  | -0.56            | (-5.60) | -0.84              | (-5.20) | -0.03          | (-2.20) |
| 3-year                     | 1.19  | 3.16  | -0.97            | (-5.70) | -1.44              | (-5.26) | -0.05          | (-2.24) |
| 5-year                     | 1.93  | 5.86  | -1.85            | (-5.90) | -2.21              | (-4.34) | -0.13          | (-3.28) |
| 7-year                     | 2.35  | 6.97  | -2.33            | (-6.31) | -2.46              | (-4.08) | -0.18          | (-3.89) |
| 10-year                    | 2.19  | 8.29  | -2.68            | (-6.07) | -3.10              | (-4.30) | -0.19          | (-3.40) |
| 20-year                    | 2.95  | 11.32 | -4.16            | (-7.05) | -3.79              | (-3.92) | -0.35          | (-4.82) |
| 30-year                    | 2.94  | 13.29 | -5.18            | (-7.60) | -3.72              | (-3.33) | -0.51          | (-6.00) |
| <i>Agency bond</i>         |       |       |                  |         |                    |         |                |         |
| 1-5 years                  | 1.83  | 3.94  | -1.17            | (-4.99) | -1.90              | (-4.66) | -0.05          | (-2.03) |
| 5-10 years                 | 3.58  | 5.20  | -1.48            | (-3.89) | -0.26              | (-0.21) | -0.14          | (-3.70) |
| 10-15 years                | 3.62  | 8.64  | -2.84            | (-5.69) | -3.71              | (-4.25) | -0.18          | (-3.10) |
| >15 years                  | 4.76  | 10.38 | -3.42            | (-5.72) | -3.63              | (-3.44) | -0.26          | (-3.66) |
| <i>Corporate bond</i>      |       |       |                  |         |                    |         |                |         |
| 1-3 years                  | 2.26  | 3.21  | -0.48            | (-2.44) | -1.56              | (-4.69) | 0.02           | (0.70)  |
| 3-5 years                  | 2.93  | 4.89  | -0.84            | (-2.78) | -2.14              | (-4.17) | 0.00           | (0.06)  |
| 5-10 years                 | 3.61  | 6.91  | -1.25            | (-2.93) | -2.98              | (-4.05) | -0.01          | (-0.26) |
| >15 years                  | 4.27  | 10.13 | -2.85            | (-4.98) | -4.47              | (-4.66) | -0.13          | (-1.91) |
| <i>Currency</i>            |       |       |                  |         |                    |         |                |         |
| Dollar-carry               | 5.34  | 8.82  | -0.98            | (-1.52) | -4.17              | (-2.08) | 0.00           | (-0.04) |
| Carry-1                    | -1.81 | 7.94  | 0.33             | (0.57)  | -0.52              | (-0.28) | 0.06           | (0.95)  |
| Carry-2                    | -0.25 | 7.47  | 1.60             | (2.99)  | 1.72               | (1.03)  | 0.14           | (2.55)  |
| Carry-3                    | 1.12  | 7.27  | 1.02             | (1.92)  | -0.04              | (-0.02) | 0.11           | (2.02)  |
| Carry-4                    | 2.53  | 8.20  | 0.45             | (0.74)  | -2.50              | (-1.34) | 0.10           | (1.60)  |
| Carry-5                    | 3.43  | 8.76  | 1.44             | (2.28)  | -1.28              | (-0.65) | 0.19           | (2.94)  |
| Carry-6                    | 5.56  | 10.10 | 1.38             | (1.87)  | -3.62              | (-1.60) | 0.20           | (2.72)  |
| <i>Commodity</i>           |       |       |                  |         |                    |         |                |         |
| Livestock                  | 2.70  | 16.99 | 1.24             | (1.24)  | -1.09              | (-0.66) | 0.15           | (1.22)  |
| Industrial metal           | 4.23  | 25.69 | 4.73             | (2.98)  | -1.07              | (-0.39) | 0.66           | (3.66)  |
| Precious metal             | 3.41  | 20.96 | 3.28             | (2.65)  | -0.22              | (-0.11) | 0.43           | (2.96)  |
| Energy                     | 7.26  | 36.93 | 16.51            | (7.05)  | -0.76              | (-0.11) | 1.78           | (7.54)  |
| Agriculture                | 0.28  | 22.24 | 4.20             | (3.28)  | 2.06               | (0.96)  | 0.26           | (1.66)  |
| <i>REIT</i>                |       |       |                  |         |                    |         |                |         |
| Equity                     | 8.31  | 17.87 | 0.72             | (0.61)  | -6.48              | (-3.20) | 0.35           | (2.77)  |
| Mortgage                   | 4.73  | 21.15 | -2.25            | (-1.63) | -8.61              | (-3.56) | 0.04           | (0.25)  |
| Hybrid                     | 8.20  | 20.31 | -1.05            | (-0.79) | -6.14              | (-2.60) | 0.12           | (0.79)  |
| <i>International stock</i> |       |       |                  |         |                    |         |                |         |
| North America              | 6.82  | 16.36 | -0.92            | (-0.96) | -5.47              | (-3.57) | 0.23           | (2.02)  |
| Europe                     | 6.60  | 18.63 | -0.93            | (-0.85) | -6.09              | (-3.48) | 0.20           | (1.56)  |
| Far East                   | 7.01  | 22.66 | -1.33            | (-0.99) | -5.05              | (-2.32) | 0.15           | (0.93)  |

- Treasuries show increasing magnitude (more negative) of headline betas with maturity; core betas are also negative.
- Commodity portfolios (especially energy futures) have huge positive exposure to energy inflation but essentially no exposure to core inflation.

This heterogeneity matters because it creates dispersion in  $\beta_i$  that is crucial for pricing tests.

## 5 Pricing inflation risks across asset classes

### 5.1 Linear beta pricing equation (paper Eq. (3))

Standard asset pricing logic implies that expected excess returns are linear in factor exposures:

$$\mathbb{E}(re_{i,t}) = \beta_i^\top \lambda, \quad (3)$$

where  $\beta_i$  is the vector of inflation betas and  $\lambda$  is the vector of **prices of risk**. The paper estimates  $\lambda$  using a two-step (Fama–MacBeth style) procedure:

**Step 1.** Estimate  $\beta_i$  from time-series regression (2).

**Step 2.** Regress average returns  $\overline{re}_i$  on estimated  $\hat{\beta}_i$  across assets to obtain  $\hat{\lambda}$ , with Shanken-adjusted standard errors.

## 5.2 Main pricing result (Table 4)

Table 4 is the empirical centerpiece:

- With **headline inflation only**,  $\hat{\lambda}^h$  is statistically insignificant and the cross-sectional fit is near zero.
- With **core and energy** as separate factors, **core** carries a *negative* and statistically significant price of risk, while energy's price is imprecise.

|                    | A. 8 average portfolios |         | B. 38 portfolios |         |
|--------------------|-------------------------|---------|------------------|---------|
| Headline $\lambda$ | 0.14                    |         | -0.08            |         |
| <i>t</i> -stat     | (0.47)                  |         | (-0.32)          |         |
| Core $\lambda$     |                         | -1.03   |                  | -1.07   |
| <i>t</i> -stat     |                         | (-2.94) |                  | (-3.72) |
| Energy $\lambda$   |                         | 3.86    |                  | 3.81    |
| <i>t</i> -stat     |                         | (1.35)  |                  | (1.36)  |
| $R^2$              | .01                     | .99     | .00              | .92     |

|                    | Stock   | Treasury | Aggr    | Corp    | Curr    | Comm    | REIT    | Intl    | Aver    | All     |
|--------------------|---------|----------|---------|---------|---------|---------|---------|---------|---------|---------|
| <i>A. Core</i>     |         |          |         |         |         |         |         |         |         |         |
| Mean               | -1.26   | -0.86    | -0.68   | -1.05   | -1.13   | -1.38   | -1.05   | -0.97   | -0.91   | -0.99   |
| <i>t</i> -stat     | (-3.31) | (-2.84)  | (-2.09) | (-3.06) | (-3.92) | (-1.16) | (-3.25) | (-2.09) | (-2.92) | (-3.61) |
| Sharpe ratio       | -0.44   | -0.36    | -0.27   | -0.49   | -0.64   | -0.17   | -0.51   | -0.31   | -0.40   | -0.49   |
| <i>B. Energy</i>   |         |          |         |         |         |         |         |         |         |         |
| Mean               | 2.02    | 0.64     | -8.25   | 6.66    | 1.34    | 12.73   | 3.47    | 8.08    | 5.23    | 5.71    |
| <i>t</i> -stat     | (0.61)  | (0.19)   | (-1.30) | (2.07)  | (0.18)  | (1.88)  | (0.55)  | (1.58)  | (2.03)  | (2.10)  |
| Sharpe ratio       | 0.09    | 0.03     | -0.18   | 0.30    | 0.03    | 0.36    | 0.09    | 0.24    | 0.28    | 0.29    |
| <i>C. Headline</i> |         |          |         |         |         |         |         |         |         |         |
| Mean               | -2.81   | -0.80    | -1.39   | -1.40   | 0.79    | 1.07    | 0.89    | -2.92   | 0.13    | -0.11   |
| <i>t</i> -stat     | (-3.36) | (-2.24)  | (-3.07) | (-2.85) | (0.88)  | (1.61)  | (1.12)  | (-2.34) | (0.42)  | (-0.35) |
| Sharpe ratio       | -0.45   | -0.30    | -0.46   | -0.42   | 0.17    | 0.29    | 0.18    | -0.34   | 0.06    | -0.05   |

The table reports the characteristics of Fama-MacBeth factor-mimicking portfolios for core, energy, and headline inflation. Panels A and B use core and energy inflation as risk factors. Panel C uses headline inflation as the single risk factor. The table reports the mean, the *t*-statistics, and the Sharpe ratio of these portfolios. The columns indicate the test assets used to construct the mimicking portfolios. Returns are annualized. Data are quarterly from 1963Q3 to 2019Q4.

**Economic meaning of a negative core risk price.** Most assets have  $\beta_i^c < 0$  (they fall when core inflation is unexpectedly high). If  $\lambda^c < 0$ , then  $\beta_i^c \lambda^c > 0$  so assets with *more negative* core betas earn *higher* average returns. In this logic, **core inflation is a bad state** for investors.

## 5.3 Visual confirmation (Figures 2 and 3)

- Figure 2: a headline-only model badly misprices the cross section; a core+energy model aligns predicted and realized returns closely.
- Figure 3: across 38 portfolios, average returns line up strongly with core betas (negative relation), much more so than with headline betas.

## 6 Factor-mimicking portfolios and within-class consistency

### 6.1 Why build mimicking portfolios?

Core and energy inflation shocks are not directly traded factors. A factor-mimicking portfolio translates the pricing information into the space of traded returns.

### 6.2 Construction (paper's Fama-MacBeth recipe)

Let  $B$  be the  $N \times K$  matrix of estimated betas across  $N$  test assets and  $K$  factors. The paper constructs mimicking-portfolio weights as

$$W \equiv (B^\top B)^{-1} B^\top, \quad (4)$$



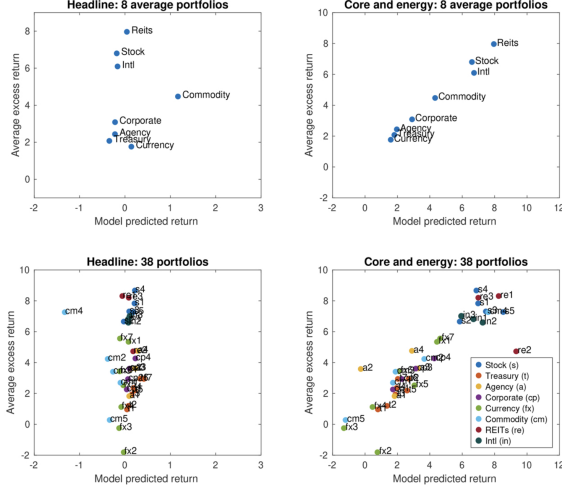


Figure 2  
Average returns versus model predicted returns: Headline inflation

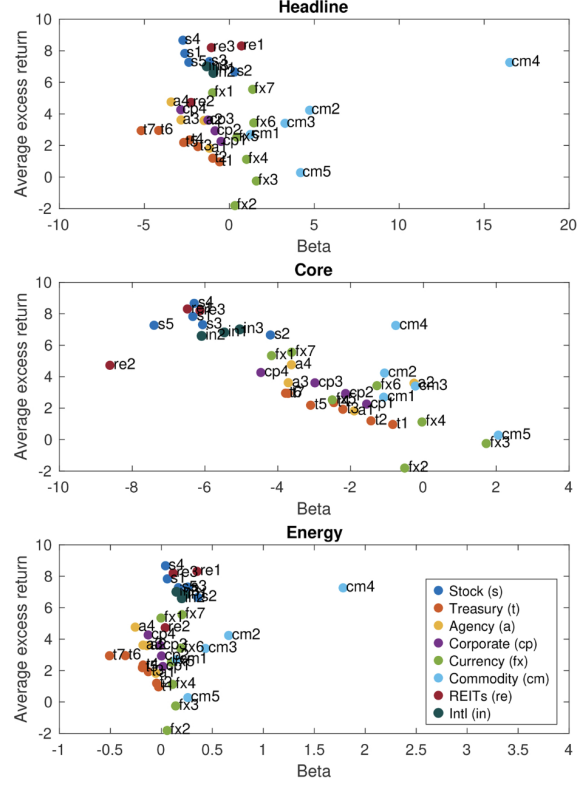


Figure 3  
Headline inflation betas and average returns

and defines factor-mimicking returns as

$$f_t^{MP} \equiv W re_t, \quad (5)$$

where  $re_t$  is the  $N \times 1$  vector of test-asset excess returns. By construction,  $WB = I_K$ , so the  $k$ th mimicking portfolio has *unit exposure* to factor  $k$  and is orthogonal to other factors in the beta space.

### 6.3 Interpreting Table 5

Table 5 reports mean returns and Sharpe ratios of mimicking portfolios constructed: (i) within each asset class and (ii) using the pooled cross section. The striking finding is:

- **Core mimicking portfolios:** consistently negative average returns around  $\approx -1\%$  per year across asset classes.
- **Energy (and headline) mimicking portfolios:** unstable across classes (sign flips, weak significance).

This is a powerful internal consistency check: the core inflation premium is *stable* across trading universes.

## 7 Time variation: betas, prices of risk, and stock–bond correlation

### 7.1 Subsample evidence (Table 6)

Table 6 splits the sample into 1963–1999 and 2000–2019 and reports:

- (1) **Betas:** core betas are comparatively stable; headline and energy betas shift materially.
- (2) **Prices of risk:** the core price of risk remains negative and significant in both subsamples; energy’s price is not stable.
- (3) **Macro comovement:** inflation–output correlations change sign (notably for energy inflation), supporting a changing cyclical role of energy inflation.

### 7.2 Conditional prices of risk (time-varying $\lambda_t$ )

The paper also allows time-varying risk prices using an affine specification:

$$\frac{M_{t+1} - \mathbb{E}_t[M_{t+1}]}{\mathbb{E}_t[M_{t+1}]} = -\lambda_t^\top \varepsilon_{\pi,t+1}, \quad (6)$$

with  $\lambda_t$  specified as a linear function of a conditioning variable  $F_t$  (the term spread), estimated using the Adrian–Crump–Moench (2015) procedure. The qualitative takeaway is that core risk pricing is relatively stable, while energy-related dynamics are more time-varying.

### 7.3 Connecting to stock–bond correlation

A coherent narrative that links this section to the earlier results is:

Core inflation is consistently “bad” for most assets and is stably priced; energy inflation is volatile and its cyclical changes, which shifts stock and bond exposures and hence changes their correlation over time.

## 8 Are core and energy inflation risks spanned by other macro factors?

### 8.1 Motivation

If core inflation risk is simply a proxy for another known macro risk (consumption growth, industrial production, intermediary constraints, CAPM, etc.), then controlling for those factors should eliminate its price of risk.

### 8.2 Empirical spanning tests (Table 7)

Table 7 adds macro factors one-by-one (and, when relevant, in pairs) in the pricing regression. The robust conclusion is:

- the estimated **core** price of risk stays around  $\approx -1$  and remains statistically significant;

**Table 6**  
Subsample Analysis

|                                 | A. Headline      |         | B. Core and energy |         |                |         | C. Test break p-value |      |        |
|---------------------------------|------------------|---------|--------------------|---------|----------------|---------|-----------------------|------|--------|
|                                 | headline $\beta$ | t-stat  | core $\beta$       | t-stat  | energy $\beta$ | t-stat  | headline              | core | energy |
| 1963–1999                       |                  |         |                    |         |                |         |                       |      |        |
| Stock                           | −5.42            | (−4.20) | −5.19              | (−3.26) | −0.24          | (−1.01) |                       |      |        |
| Treasury                        | −2.88            | (−5.52) | −2.77              | (−4.31) | −0.20          | (−2.03) |                       |      |        |
| Agency                          | −2.49            | (−4.31) | −2.50              | (−3.56) | −0.12          | (−1.20) |                       |      |        |
| Corporate                       | −3.23            | (−5.76) | −3.08              | (−4.44) | −0.22          | (−2.22) |                       |      |        |
| Currency                        | 0.12             | (0.08)  | 0.26               | (0.09)  | 0.06           | (0.33)  |                       |      |        |
| Commodity                       | 4.55             | (2.80)  | 0.42               | (0.21)  | 0.65           | (2.24)  |                       |      |        |
| REIT                            | −5.79            | (−3.76) | −5.91              | (−3.29) | −0.37          | (−1.43) |                       |      |        |
| Intl stock                      | −5.92            | (−4.60) | −5.12              | (−3.24) | −0.46          | (−1.97) |                       |      |        |
| 2000–2019                       |                  |         |                    |         |                |         |                       |      |        |
| Stock                           | 2.96             | (2.22)  | −6.30              | (−1.18) | 0.35           | (2.63)  | 0.00                  | 0.84 | 0.03   |
| Treasury                        | −2.23            | (−4.73) | −0.29              | (−0.15) | −0.22          | (−4.65) | 0.36                  | 0.26 | 0.83   |
| Agency                          | −1.05            | (−4.38) | −0.27              | (−0.27) | −0.10          | (−4.11) | 0.02                  | 0.18 | 0.82   |
| Corporate                       | −0.20            | (−0.49) | −0.66              | (−0.40) | −0.01          | (−0.22) | 0.00                  | 0.24 | 0.04   |
| Currency                        | 1.30             | (2.67)  | −1.87              | (−0.96) | 0.15           | (3.10)  | 0.37                  | 0.52 | 0.55   |
| Commodity                       | 12.44            | (8.35)  | −0.38              | (−0.06) | 1.24           | (8.44)  | 0.00                  | 0.90 | 0.07   |
| REIT                            | 3.47             | (2.22)  | −10.18             | (−1.65) | 0.44           | (2.88)  | 0.00                  | 0.47 | 0.02   |
| Intl stock                      | 3.28             | (2.47)  | −6.62              | (−1.25) | 0.39           | (2.95)  | 0.00                  | 0.78 | 0.00   |
| D. Price of risk                |                  |         |                    |         |                |         |                       |      |        |
|                                 | 1963–1999        |         |                    |         | 2000–2019      |         |                       |      |        |
| Headline $\lambda$              | −0.25            |         |                    |         | 0.35           |         |                       |      |        |
| t-stat                          | (−0.93)          |         |                    |         | (0.72)         |         |                       |      |        |
| Core $\lambda$                  |                  |         |                    |         | −0.78          |         |                       |      |        |
| t-stat                          |                  |         |                    |         | (−2.02)        |         |                       |      |        |
| Energy $\lambda$                |                  |         |                    |         | 3.12           |         |                       |      |        |
| t-stat                          |                  |         |                    |         | (1.39)         |         |                       |      |        |
| $R^2$                           | 0.07             |         |                    |         | 0.06           |         |                       |      |        |
|                                 | 0.57             |         |                    |         | 0.44           |         |                       |      |        |
| E. Inflation-output correlation |                  |         |                    |         |                |         |                       |      |        |
|                                 | 1963–1999        |         |                    |         | 2000–2019      |         |                       |      |        |
| Headline                        | −0.25            |         |                    |         | 0.33           |         |                       |      |        |
| Core                            | −0.31            |         |                    |         | 0.04           |         |                       |      |        |
| Energy                          | −0.08            |         |                    |         | 0.35           |         |                       |      |        |

**Table 7**  
Price of inflation risks with macroeconomic factors

|        | Cons    | Cons/Dur | IP      | Pay     | Unem    | HHL     | Unf Cons | Cap     | Itmdr   | CAPM    |
|--------|---------|----------|---------|---------|---------|---------|----------|---------|---------|---------|
| Core   | -1.06   | -1.04    | -1.07   | -1.07   | -1.06   | -1.04   | -1.07    | -1.08   | -1.05   | -1.06   |
| t-stat | (-3.69) | (-3.67)  | (-3.51) | (-3.27) | (-3.39) | (-3.48) | (-3.70)  | (-3.72) | (-3.65) | (-3.39) |
| Energy | 3.90    | 4.38     | 4.08    | 3.68    | 3.84    | 3.97    | 3.98     | 3.94    | 4.50    | 3.83    |
| t-stat | (1.29)  | (1.36)   | (1.38)  | (1.33)  | (1.36)  | (1.29)  | (1.44)   | (1.38)  | (1.58)  | (1.35)  |
| Macro  | 0.10    | 0.17     | -0.34   | -0.08   | 0.11    | 0.46    | 0.00     | -0.31   | 2.03    | 6.78    |
| t-stat | (0.18)  | (0.32)   | (-0.24) | (-0.16) | (0.26)  | (0.62)  | (0.26)   | (-0.59) | (1.76)  | (2.93)  |
| Macro2 |         | -2.62    |         |         |         | -0.01   |          |         |         |         |
| t-stat |         | (-0.67)  |         |         |         | (-0.58) |          |         |         |         |
| $R^2$  | 0.92    | 0.92     | 0.92    | 0.92    | 0.92    | 0.93    | 0.93     | 0.92    | 0.92    | 0.92    |

The table reports the prices of risks in various specifications using 38 test portfolios as test assets. The macroeconomic factors include consumption growth (Cons), durable consumption growth (Cons/Dur), industrial production growth (IP), payroll growth (Pay), unemployment rate growth (Unem), long-run consumption growth rate and short-run consumption growth news (HHL), unfiltered consumption growth (Unf Cons), and capital share growth (Cap), intermediary factor (Itmdr), and the aggregate stock market return (CAPM). Prices of risks are estimated using a two-step procedure and the  $t$ -statistics are calculated using Shanken-adjusted standard errors. The second-step  $R^2$  is reported in the last row. Data are quarterly from 1963Q3 to 2019Q4.

- macro factor prices are generally insignificant in this broad cross-asset test;
- cross-sectional  $R^2$  remains high.

Thus, core inflation risk is not easily “explained away” by standard macro factors.

## 9 Mechanisms: growth, cash flows vs discount rates, and monetary policy

### 9.1 Inflation and real growth predictability (Table 8)

Table 8 regresses future real growth (GDP, consumption, dividends) on current inflation. The key mechanism insight is:

- headline inflation predicts lower future growth,
- but the predictive content is **concentrated in core inflation**;
- energy inflation has weak or insignificant predictive power for real growth.

This aligns with the asset-pricing result: core inflation is the component that covaries systematically with long-run real fundamentals.

### 9.2 High-frequency inflation announcement evidence (Table 9 and Figure 4)

The paper uses a tight intraday window around inflation announcements and defines inflation surprises as (actual minus survey-median forecast).

**Table 8**  
Inflation and real growth

|             | headline | <i>t</i> -stat | <i>R</i> <sup>2</sup> | core  | <i>t</i> -stat | energy | <i>t</i> -stat | <i>R</i> <sup>2</sup> |
|-------------|----------|----------------|-----------------------|-------|----------------|--------|----------------|-----------------------|
| 1 quarter   |          |                |                       |       |                |        |                |                       |
| GDP         | -0.14    | (-1.21)        | 0.02                  | -0.21 | (-1.88)        | 0.00   | (-0.23)        | 0.03                  |
| Consumption | -0.22    | (-2.42)        | 0.08                  | -0.22 | (-2.32)        | -0.01  | (-0.86)        | 0.07                  |
| Dividend    | -0.27    | (-1.15)        | 0.02                  | -0.67 | (-4.27)        | 0.04   | (0.96)         | 0.06                  |
| 1 year      |          |                |                       |       |                |        |                |                       |
| GDP         | -0.75    | (-2.34)        | 0.08                  | -0.70 | (-2.24)        | -0.05  | (-1.05)        | 0.07                  |
| Consumption | -0.66    | (-2.24)        | 0.09                  | -0.46 | (-1.81)        | -0.05  | (-1.12)        | 0.05                  |
| Dividend    | -1.26    | (-1.12)        | 0.03                  | -2.93 | (-5.78)        | 0.18   | (0.95)         | 0.11                  |

The table reports the results of regressions of future real growth on current headline, core, and energy inflation at one-quarter and 1-year horizons. The *t*-statistics are in parentheses. Variables are annualized. Data are quarterly from 1963Q3 to 2019Q4.

**Table 9**  
Inflation announcement event study

| A. Fed funds futures         |       |                |          |                |       |                |         |
|------------------------------|-------|----------------|----------|----------------|-------|----------------|---------|
|                              | core  | <i>t</i> -stat | headline | <i>t</i> -stat |       |                |         |
| (1)                          | 3.17  | (5.55)         |          |                |       |                |         |
| (2)                          |       |                | 1.65     | (3.38)         |       |                |         |
| (3)                          | 3.04  | (4.32)         | 0.19     | (0.32)         |       |                |         |
| B. Two-year Treasury futures |       |                |          |                |       |                |         |
|                              | core  | <i>t</i> -stat | headline | <i>t</i> -stat |       |                |         |
| (1)                          | 0.29  | (5.24)         |          |                |       |                |         |
| (2)                          |       |                | 0.16     | (3.39)         |       |                |         |
| (3)                          | 0.27  | (3.94)         | 0.03     | (0.50)         |       |                |         |
| C. Stock futures             |       |                |          |                |       |                |         |
|                              | core  | <i>t</i> -stat | headline | <i>t</i> -stat | ff    | <i>t</i> -stat | ty      |
| (1)                          | -2.14 | (-8.84)        |          |                |       |                |         |
| (2)                          |       |                | -1.53    | (-7.40)        |       |                |         |
| (3)                          | -1.61 | (-5.49)        | -0.76    | (-3.13)        |       |                |         |
| (4)                          |       |                |          |                | -0.27 | (-11.15)       |         |
| (5)                          | -1.42 | (-6.03)        |          |                | -0.22 | (-9.29)        |         |
| (6)                          | -1.29 | (-5.29)        |          |                | -0.21 | (-7.87)        | -1.24   |
|                              |       |                |          |                |       |                | (-4.39) |

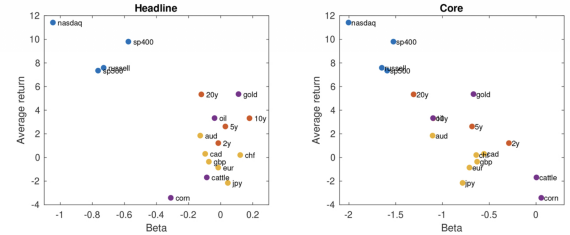
**Policy-rate expectations.** Panels A–B of Table 9 show that positive **core** surprises raise expected policy rates (Fed funds futures and 2-year Treasury futures), whereas headline surprises add little once core surprises are controlled for.

**Equity response and decomposition.** Panel C shows equity index futures fall strongly with core surprises. Including futures-based policy-rate changes as controls reduces but does not eliminate the equity response, suggesting that monetary policy is an important channel but not the only channel.

**Figure 4 as a cross-asset-class check.** Figure 4 plots average returns against announcement-window inflation betas and visually confirms that core-related exposures line up with average returns even in high-frequency data.

**Table 10**  
Flexible versus sticky inflation

| A. Asset return exposures |                      |                |               |                |
|---------------------------|----------------------|----------------|---------------|----------------|
|                           | sticky               | <i>t</i> -stat | flexible      | <i>t</i> -stat |
| Stock                     | -4.68                | (-2.99)        | 0.25          | (0.61)         |
| Treasury                  | -1.12                | (-1.86)        | -0.93         | (-5.93)        |
| Agency                    | -0.94                | (-1.93)        | -0.51         | (-4.20)        |
| Corporate                 | -1.61                | (-2.70)        | -0.39         | (-2.56)        |
| Currency                  | -1.14                | (-0.69)        | 0.41          | (2.16)         |
| Commodity                 | -1.53                | (-0.87)        | 3.88          | (8.51)         |
| REIT                      | -4.35                | (-2.38)        | 0.61          | (1.38)         |
| Intl stock                | -4.95                | (-3.27)        | 0.23          | (0.58)         |
| B. Price of risk          |                      |                |               |                |
|                           | 8 average portfolios |                | 38 portfolios |                |
| Sticky $\lambda$          | -1.50                |                | -1.45         |                |
| <i>t</i> -stat            | (-2.61)              |                | (-3.49)       |                |
| Flexible $\lambda$        | 0.45                 |                | -0.21         |                |
| <i>t</i> -stat            | (0.47)               |                | (-0.24)       |                |
| <i>R</i> <sup>2</sup>     | .96                  |                | .89           |                |



**Figure 4**  
Average returns and inflation announcement betas  
The figure plots the average excess return of the 18 futures returns against their headline and core inflation betas. The horizontal axis shows headline inflation betas and the vertical axis shows the average excess returns. Each dot represents a test asset, and different colors refer to assets from different asset classes. Betas are estimated using the event-window inflation announcement return. Average return are estimated using all the days.

## 10 Sticky versus flexible inflation and the economic mechanism

### 10.1 Motivation

Core inflation is largely driven by categories with sticky prices; energy inflation is largely flexible. If the pricing results are fundamentally about price stickiness, then re-labelling inflation factors as “sticky” vs “flexible” should replicate the core vs energy results.

## 10.2 Evidence (Table 10)

Table 10 confirms:

- **Sticky inflation** has a negative and significant price of risk and assets load negatively on it (especially stocks and REITs).
- **Flexible inflation** behaves more like energy: commodities hedge it, and its risk price is not robustly significant.

This empirical pattern directly motivates the two-sector NK model in the next section.

## 11 Two-sector New Keynesian asset pricing model

### 11.1 Model goal

The structural model is designed to explain *jointly*:

- (1) why core (sticky) inflation is the priced inflation risk,
- (2) why many real assets hedge energy inflation but not core inflation,
- (3) why stock–bond correlation changes when energy inflation becomes more procyclical and volatile.

### 11.2 Households and preferences (paper Eqs. (3)–(7))

The representative household has recursive (Epstein–Zin) preferences:

$$U_t = \left[ (1 - \delta) \tilde{C}_t^{1 - \frac{1}{\psi}} + \delta \mathbb{E}_t \left( U_{t+1}^{1 - \gamma} \right)^{\frac{1 - \frac{1}{\psi}}{1 - \gamma}} \right]^{\frac{1}{1 - \frac{1}{\psi}}}, \quad (7)$$

where  $\psi$  is the intertemporal elasticity of substitution and  $\gamma$  is risk aversion.

The composite  $\tilde{C}_t$  aggregates consumption and leisure:

$$\tilde{C}_t = \left[ \frac{1}{1 + \omega} C_t^{\frac{\xi_t - 1}{\xi_t}} + \frac{\omega}{1 + \omega} \left( N_{t-1} (\bar{L} - L_t) \right)^{\frac{\xi_t - 1}{\xi_t}} \right]^{\frac{\xi_t}{\xi_t - 1}}, \quad (8)$$

where  $\bar{L}$  is total labor endowment and  $L_t$  is labor supply (so  $\bar{L} - L_t$  is leisure), scaled by trend  $N_{t-1}$ .

The consumption bundle  $C_t$  is a CES aggregate of core and energy goods:

$$C_t = \left[ \alpha_c C_{c,t}^{\frac{\phi - 1}{\phi}} + (1 - \alpha_c) \left( e^{\eta_t} C_{e,t} \right)^{\frac{\phi - 1}{\phi}} \right]^{\frac{\phi}{\phi - 1}}, \quad (9)$$

where  $\phi$  is the elasticity of substitution and  $\eta_t$  is a relative energy demand shifter.

Let  $P_{e,t}$  denote the real energy price in units of core goods. Optimal consumption choice implies:

$$P_{e,t} = \frac{1 - \alpha_c}{\alpha_c} \exp(\eta_t)^{1 - \frac{1}{\phi}} \left( \frac{C_{c,t}}{C_{e,t}} \right)^{\frac{1}{\phi}}. \quad (10)$$

The (core-goods) stochastic discount factor is:

$$M_{t+1} = \delta \left( \frac{C_{t+1}}{C_t} \right)^{-\frac{1}{\xi_l}} \left( \frac{\tilde{C}_{t+1}}{\tilde{C}_t} \right)^{\frac{1}{\xi_l} - \frac{1}{\psi}} \left( \frac{U_{t+1}^{1-\gamma}}{\mathbb{E}_t[U_{t+1}^{1-\gamma}]} \right)^{1 - \frac{1}{\theta}} \frac{\partial C_{t+1} / \partial C_{c,t+1}}{\partial C_t / \partial C_{c,t}}, \quad \theta \equiv \frac{1 - \gamma}{1 - \frac{1}{\psi}}. \quad (11)$$

### 11.3 Core sector production and price stickiness (paper Eqs. (8)–(12))

Final core output aggregates intermediate varieties:

$$Y_t = \left( \int_0^1 X_{i,t}^{\frac{\nu-1}{\nu}} di \right)^{\frac{\nu}{\nu-1}}, \quad (12)$$

with elasticity of substitution  $\nu$ . Profit maximization implies intermediate-good demand:

$$X_{i,t} = Y_t P_{i,t}^{-\nu}. \quad (13)$$

Each intermediate producer has Cobb–Douglas technology:

$$X_{i,t} = K_{i,t}^\alpha (Z_t L_{i,t})^{1-\alpha}, \quad (14)$$

with productivity  $Z_t$  (trend and transitory components, with stochastic volatility as in the paper).

Capital accumulates with adjustment costs:

$$K_{i,t+1} = (1 - \delta_k) K_{i,t} + \Phi_K \left( \frac{I_{i,t}}{K_{i,t}} \right) K_{i,t}, \quad (15)$$

where the paper parameterizes:

$$\Phi_K \left( \frac{I_{i,t}}{K_{i,t}} \right) = \frac{a_{1,k}}{1 - \frac{1}{\xi_k}} \left( \frac{I_{i,t}}{K_{i,t}} \right)^{1 - \frac{1}{\xi_k}} + a_{2,k}. \quad (16)$$

Intermediate producers face sticky prices via a convex (Rotemberg-type) price adjustment cost (given in the paper).

### 11.4 Energy sector and monetary policy (paper Eqs. (13)–(15))

Energy is endowed each period; its demand and supply evolve exogenously:

$$\eta_t = \rho_\eta \eta_{t-1} + \sigma_\eta \varepsilon_{\eta,t} + \delta_a \varepsilon_{n,t}, \quad \log \left( \frac{C_{e,t}}{N_{t-1}} \right) = (1 - \rho_e) \log(C_{e,ss}) + \rho_e \log \left( \frac{C_{e,t-1}}{N_{t-2}} \right) + \sigma_e \varepsilon_{e,t}.$$

Monetary policy follows a Taylor rule responding to (core) inflation and detrended output, as in the paper.

### 11.5 Asset prices (paper Eq. (16) and Euler equations)

Aggregate dividends combine core and energy payouts:

$$D_t^{agg} = \int_0^1 D_{i,t} di + \alpha C_{e,t} P_{e,t},$$

and asset prices satisfy standard Euler equations, e.g.  $\mathbb{E}_t[M_{t+1}R_{t+1}] = 1$ .

## 12 Quantitative results and interpretation of the model tables

### 12.1 Main model fit (Table 11)

Table 11 compares data moments to simulated model moments. A coherent read is to go panel-by-panel:

- **Panel A (macro moments):** the model matches the order of magnitude of consumption growth volatility and energy-growth volatility, which is crucial because energy-demand shocks drive time-varying comovement.
- **Panel B (inflation moments):** energy inflation remains highly volatile and weakly persistent; core inflation is smoother and persistent.
- **Panels C–D (returns and betas):** the model reproduces the negative stock core beta and positive stock energy beta, and negative bond betas, consistent with the empirical regressions.
- **Panel E (risk prices):** the model matches the negative core inflation price of risk remarkably closely.

### 12.2 Parameter sensitivity (Table 12)

Table 12 varies key parameters (price stickiness, market power, substitution elasticities, persistence, and policy aggressiveness). The message is not that every moment is perfectly stable, but that the qualitative mechanism is robust:

- weakening core price stickiness changes the core beta and the core risk price,
- altering substitution elasticities shifts inflation volatilities and betas,
- more aggressive monetary policy affects core inflation volatility and bond exposures.

**Table 11**  
**Main quantitative results**

| Data                                   |       |             |          | Model |             |          |
|----------------------------------------|-------|-------------|----------|-------|-------------|----------|
| <i>A. Macroeconomic moments</i>        |       |             |          |       |             |          |
| $E[\Delta y]$                          | 1.96  |             |          | 1.78  |             |          |
| $\sigma(\Delta y)$                     | 3.18  |             |          | 2.67  |             |          |
| $\sigma(\Delta c)$                     | 1.74  |             |          | 1.37  |             |          |
| $\sigma(\Delta i)$                     | 15.15 |             |          | 8.06  |             |          |
| $\sigma(\Delta c_e)$                   | 4.00  |             |          | 4.18  |             |          |
| $E[r_f]$                               | 0.93  |             |          | 1.45  |             |          |
| $E[y^{(1)}]$                           | 4.63  |             |          | 4.98  |             |          |
| <i>B. Inflation</i>                    |       |             |          |       |             |          |
|                                        | Core  | Energy      | Headline | Core  | Energy      | Headline |
| Mean                                   | 3.75  | 3.75        | 3.76     | 3.61  | 3.60        | 3.61     |
| SD                                     | 2.66  | 19.52       | 3.24     | 3.29  | 19.48       | 3.92     |
| Autocorr                               | 0.79  | 0.04        | 0.6      | 0.94  | 0.01        | 0.65     |
| <i>C. Asset excess returns</i>         |       |             |          |       |             |          |
|                                        | Mean  | <i>S.D.</i> |          | Mean  | <i>S.D.</i> |          |
| Stock                                  | 6.80  | 16.79       |          | 6.87  | 18.58       |          |
| Bond                                   | 1.93  | 5.86        |          | 1.13  | 2.62        |          |
| Currency                               | 1.76  | 7.05        |          | 0.25  | 7.28        |          |
| Commodity                              | 4.47  | 21.90       |          | 1.88  | 9.72        |          |
| <i>D. Exposures to inflation risks</i> |       |             |          |       |             |          |
|                                        | Core  | Energy      | Headline | Core  | Energy      | Headline |
| Stock                                  | -5.60 | 0.21        | -1.33    | -5.70 | 0.34        | 1.07     |
| Bond                                   | -2.21 | -0.13       | -1.85    | -1.27 | -0.08       | -0.87    |
| Currency                               | -1.04 | 0.13        | 1.04     | -0.89 | 0.02        | -0.05    |
| Commodity                              | -0.07 | 1.10        | 8.59     | -0.10 | 1.00        | 6.84     |
| <i>E. Prices of inflation risks</i>    |       |             |          |       |             |          |
| $\lambda$                              | -1.03 | 3.86        | 0.14     | -1.07 | 1.80        | 0.39     |

Moments are computed from quarterly data from 1963 to 2019. Returns and growth rates are annualized. Moments in the model are computed based on a long simulation of 20,000 quarters.

**Table 12**  
**Parameter sensitivity**

|                      | Benchmark | (1)<br>$\phi_p = 30$ | (2)<br>$v = 3$ | (3)<br>$\phi = 0.8$ | (4)<br>$\rho_n = 0.9$ | (5)<br>$\rho_\pi = 2$ |
|----------------------|-----------|----------------------|----------------|---------------------|-----------------------|-----------------------|
| Core inflation vol   | 3.29      | 3.29                 | 2.87           | 5.24                | 3.75                  | 1.67                  |
| Energy inflation vol | 19.48     | 19.49                | 19.48          | 22.07               | 19.61                 | 19.27                 |
| Stock excess return  | 6.87      | 6.89                 | 7.33           | 6.74                | 4.75                  | 7.22                  |
| Bond excess return   | 1.13      | 1.23                 | 0.72           | 1.70                | 1.04                  | 0.23                  |
| Stock core beta      | -5.70     | -2.51                | -14.32         | -4.93               | -3.85                 | -3.82                 |
| Bond core beta       | -1.27     | -0.57                | -1.12          | -2.41               | -1.53                 | 1.34                  |
| Stock energy beta    | 0.34      | 0.33                 | 0.40           | 0.01                | 0.28                  | 0.33                  |
| Bond energy beta     | -0.08     | -0.08                | -0.09          | 0.26                | -0.06                 | -0.06                 |
| Core price of risk   | -1.07     | -2.41                | -0.46          | -1.25               | -1.02                 | -1.42                 |
| Energy price of risk | 1.80      | 1.50                 | 1.90           | -1.11               | 1.46                  | 2.16                  |

Moments in the model are computed based on a long simulation of 20,000 quarters.



**Table 13**  
**Stock-bond correlation**

|                        | Data   |         | Model |       |       |       |       |       |       |
|------------------------|--------|---------|-------|-------|-------|-------|-------|-------|-------|
|                        | Pre-99 | Post-99 | (1)   | (2)   | (3)   | (4)   | (5)   | (6)   | (7)   |
| Stock-bond correlation | 0.28   | −0.57   | 0.47  | 0.26  | 0.08  | 0.05  | 0.00  | −0.04 | −0.31 |
| Stock core beta        | −5.19  | −6.30   | −6.22 | −4.13 | −2.35 | −0.63 | −8.06 | −1.66 | 3.09  |
| Bond core beta         | −2.77  | −0.29   | −1.42 | −1.31 | 1.23  | 0.50  | 2.95  | 1.36  | 1.76  |
| Stock energy beta      | −0.24  | 0.35    | −0.07 | 0.32  | 0.31  | 0.31  | 0.35  | 0.34  | 0.28  |
| Bond energy beta       | −0.20  | −0.22   | −0.01 | −0.05 | −0.05 | −0.05 | −0.06 | −0.06 | −0.11 |
| Core vol               | 2.85   | 0.64    | 3.21  | 3.33  | 1.69  | 1.86  | 1.51  | 1.64  | 1.53  |
| Energy vol             | 12.63  | 28.20   | 12.70 | 30.14 | 30.00 | 30.02 | 30.07 | 30.08 | 30.04 |
| Energy supply          |        |         | Y     |       |       |       |       |       |       |
| Procyclical $\eta$     |        |         |       | Y     | Y     | Y     | Y     | Y     | Y     |
| Aggressive MP          |        |         |       |       | Y     | Y     | Y     | Y     | Y     |
| Low price stickiness   |        |         |       |       |       | Y     |       | Y     | Y     |
| High market power      |        |         |       |       |       |       | Y     | Y     | Y     |
| Low transitory shock   |        |         |       |       |       |       |       |       | Y     |

Moments are computed from quarterly data from 1963 to 2019. Returns and growth rates are annualized. Moments in the model are computed based on a long simulation of 20,000 quarters. In column 1,  $\sigma_\epsilon = 0.16$ ,  $\sigma_\eta = 0$ , and  $\delta_a = 0$ . “Procyclical  $\eta$ ” set  $\delta_a$  to 0.05 and  $\sigma_\eta$  to 0.0794. “Aggressive MP” sets  $\rho_\pi$  to 2. “Low price stickiness” sets  $\phi_p$  to 30. “High market power” sets  $v$  to 3. “Low transitory shock” sets  $\varphi_a$  to 2/3.

## 12.3 Stock–bond correlation (Table 13)

Table 13 uses the model to rationalize the sign change in stock–bond correlation:

- making energy demand more volatile and procyclical pushes stock and bond returns in different directions, lowering their correlation;
- a more hawkish policy response affects bond exposures to core inflation and can further alter correlation.

This ties back to the empirical subsample evidence: energy inflation becomes more procyclical in the later sample.

# 13 Summary of what the paper establishes

## 13.1 Empirical facts

- (1) Core inflation is the inflation component to which most assets are negatively exposed.
- (2) The cross-sectional *price* of inflation risk is concentrated in core (sticky) inflation, not energy.
- (3) Energy inflation matters for hedging and time variation, but its premium is not stable.
- (4) Core inflation predicts lower future real growth, and core surprises move policy-rate expectations.

## 13.2 Structural interpretation

A two-sector NK model with sticky core prices, flexible energy prices, recursive preferences, and energy demand/supply shocks can quantitatively match the core-vs-energy pricing facts and the time variation in stock–bond correlation.

## 14 Conclusion

### 14.1 Summary

This paper concludes that **inflation risk is not a single object for asset pricing and hedging**. Instead, **decomposing headline inflation into core and energy** fundamentally changes both:

(i) which assets hedge inflation, and (ii) which inflation risk is *priced* in the cross section of returns.

The paper establishes a tight chain of empirical and theoretical results:

- (i) **Inflation betas differ sharply across components.** In quarterly data (1963Q3–2019Q4), the same asset can hedge one inflation component and be exposed negatively to another:
  - Stocks and REITs have **negative core inflation betas** and **positive energy inflation betas** (e.g., Table 2: stock core beta  $\approx -5.60$  and energy beta  $\approx 0.21$ ; REIT core beta  $\approx -6.54$  and energy beta  $\approx 0.31$ ).
  - Treasuries, agency bonds, and corporate bonds load **negatively on both** core and energy inflation shocks (Table 2).
  - Currencies and commodity futures **hedge energy inflation** (positive energy betas) but have **weak or insignificant** core betas (Table 2).

Hence, the conventional “real assets hedge inflation” view is true mainly for *energy* inflation, not for *core* inflation.

- (ii) **Only core inflation carries a robust risk premium.** The cross-sectional pricing regression

$$\mathbb{E}(r_i^e) = \beta_i^\pi \lambda_\pi$$

is uninformative when inflation is measured as headline CPI: the estimated headline price of risk is statistically insignificant (Table 4). Once inflation is decomposed, **core inflation has a stable and statistically significant negative price of risk**, while **energy inflation does not**:

- Table 4 (full sample):  $\lambda_{\text{core}} \approx -1.03$  ( $t \approx -2.94$ ) using 8 cross-asset average portfolios, and  $\lambda_{\text{core}} \approx -1.07$  ( $t \approx -3.72$ ) using 38 test portfolios.
- The energy price of risk is positive but statistically weak (Table 4:  $t \approx 1.35$ – $1.36$ ).
- The cross-sectional fit improves dramatically when using core+energy rather than headline (Table 4:  $R^2 \approx 0.99$  with 8 averages and  $\approx 0.92$  with 38 portfolios).

Economically, assets with higher average returns are precisely those that are **more negatively exposed to core inflation** (i.e., they lose when core inflation rises), so they require compensation because they perform poorly in “bad” core-inflation states.

- (iii) **The core premium is consistent within and across asset classes and is hard to span.** The estimated core premium is similar when pricing is done within each asset class or jointly across asset classes, and it remains significant after adding standard macro risk factors (spanning tests).

- (iv) **Energy inflation explains time variation in stock–bond comovement.** Energy inflation becomes more procyclical and volatile in the post-2000 subsample, and the paper shows that this helps rationalize the observed time variation in stock–bond correlation. In subsamples (Table 6), the **core risk price remains negative** (about  $-1.05$  pre-2000 and  $-0.78$  post-2000, both significant), while the energy risk price is unstable and statistically weak.
- (v) **A structural mechanism matches the facts.** A two-sector New Keynesian asset pricing model with **sticky core prices**, **flexible energy prices**, and **energy demand/supply shocks** (plus recursive preferences and monetary-policy transmission) can match: (a) why core (sticky) inflation is priced, (b) why energy inflation is primarily a hedging component, and (c) why stock–bond correlation varies with the cyclical properties of energy inflation.

## 14.2 Economic intuition: why “getting to the core” changes inflation hedging and pricing

**Step 1: core and energy inflation are different economic shocks.** Energy inflation is volatile and weakly persistent; core inflation is more persistent (Table 1). This persistence makes core inflation more informative about medium/long-run real fundamentals and policy, whereas energy inflation is often short-lived and linked to flexible-price movements.

**Step 2: sticky-price (core) inflation is tightly connected to monetary policy and real activity.** Because core prices are sticky, core inflation innovations tend to reflect (or trigger) systematic monetary-policy responses and broader demand conditions. Empirically, core inflation predicts weaker future real growth and core surprises move policy-rate expectations more strongly than headline surprises. These channels make core inflation a *fundamental macro risk* that investors care about in pricing.

**Step 3: the sign of the core premium is natural.** The estimated price of core inflation risk is **negative** ( $\lambda_{\text{core}} < 0$ ). Most risky assets (stocks, REITs, many credit portfolios) have **negative** core betas ( $\beta_{\text{core}} < 0$ ), meaning they lose when core inflation rises. Therefore,

$$\mathbb{E}(r_i^e) \approx \beta_{i,\text{core}} \lambda_{\text{core}} > 0,$$

so these assets require higher average returns as compensation for performing poorly in core-inflation states. In contrast, energy-hedging exposures (commodities, some currencies) are not rewarded with a stable premium because energy inflation is not robustly priced.

**Step 4: why headline inflation is a poor single factor.** Headline inflation mechanically mixes the persistent sticky component (core) with the volatile flexible component (energy). If the two components have different betas and different risk prices, the single-factor “headline” regression averages them in a way that can mask (or even flip) both exposures and premia. This is why the headline price of inflation risk is insignificant, while the core price of risk becomes large, stable, and precisely estimated once the decomposition is used (Table 4).

### 14.3 Implications

- **Inflation hedging should be component-specific.** Many “real assets” hedge *energy* inflation but not *core* inflation. If investors fear persistent inflation and policy tightening (core), the relevant hedges are those with *positive core betas*—which are rare.
- **Inflation risk premia are about sticky inflation.** If we want an inflation factor for cross-asset pricing, the evidence points to core (or sticky) inflation as the priced component.
- **Time variation comes from changing energy cyclicalities.** When energy inflation becomes more procyclical/volatile, it can shift stock–bond comovement and the apparent inflation-hedging role of different assets.

### 14.4 Takeaway

*Across asset classes, inflation risk premia are driven by **core (sticky) inflation**: most assets load negatively on core inflation and therefore require compensation, while energy inflation is mainly a hedging component whose pricing is unstable; a two-sector sticky-vs-flexible price model with monetary-policy transmission rationalizes these facts and the time variation in stock–bond correlation.*